

often requires the re-generation of the data graph, which wastes computing resources.

[0004] To solve these and other problems, embodiments described herein expand data graphs, such as provenance chains, to include related information from one or more data sources. For example, as noted above, a manufacturing facility may provide limited information regarding an entity, such as an identifier of the entity and a shipping date of the entity. The manufacturing facility, however, has a location and one or more data sources (for example, not maintained by the manufacturing facility) may provide information regarding the location, such as, for example, weather, traffic, and socioeconomic statistics such as, for example, political status or unrest, labor laws, employee information, or the like. Thus, the systems and methods described herein identify such related information and link the information with a data graph chain to allow a user to interact with this related information. For example, by linking in weather information, a user can interact with a data graph using weather as a dimension although the original data records for an entity did not include such information. In particular, using the linked related information, a user can query for entities that experienced particular weather conditions, such as temperatures exceeding a specified threshold although such information was not directly tracked for the entities in question.

[0005] For example, one embodiment provides a system for creating a data graph. The system includes a plurality of data sources and an electronic computing device. The electronic computing device includes an electronic processor configured to generate a data graph for an entity by accessing a plurality of data records stored in a first set of the plurality of data sources. Each of the plurality of data records is explicitly linked to the entity. The data graph includes a plurality of connected nodes and each of the plurality of nodes includes at least one feature representing a dimension of the data graph. The electronic processor is also configured to infer an additional dimension for the data graph based on data included in the data graph, in response to confirming addition of the additional dimension to the data graph, add data to the data graph associated with the additional dimension by accessing data stored in a second set of the plurality of data sources, and store the data graph with the data associated with the additional dimension.

[0006] Another embodiment provides a method of creating a data graph. The method includes generating a data graph for an entity by accessing a plurality of data records stored in a first set of the plurality of data sources. Each of the plurality of data records is explicitly linked to the entity. The data graph includes a plurality of connected nodes and each of the plurality of nodes includes at least one feature representing a dimension of the